National University of Computer and Emerging Sciences, Lahore Campus

Cyber Security (Code: CY-2004)

Assignment 1 [Section BCY-3A] Fall 2026

Due Date: November 14, 2026 **Time: Latest by 23:59** **Marks: 100** **CLO: 3**

This is a group assignment, and each group can comprise at most 3 students, and its assessment involves Practical demonstration, technical report, and presentation
Reference Material: Module 8 namely Sniffing of Certified Ethical Hacket (CEH) Version 13 by EC Council , relevant course lectures, official tool documentation, and peer-reviewed research papers.

Assignment Overview

Cybersecurity professionals use specialized tools to monitor networks, analyze traffic, identify vulnerabilities, detect malicious activities, investigate incidents, and assess the security of information systems. Understanding a tool requires more than knowing its commands or features; students must be able to configure it, demonstrate its operation, evaluate its capabilities and limitations, and examine its use in published research.

In this assignment, each group must select **one cybersecurity tool**, study it in depth, simulate its operation in a controlled environment, demonstrate at least three strengths and three weaknesses, and investigate how the tool has been used in academic research.

Each group must also identify ten relevant research papers that explicitly mention or use the selected tool, present them in a structured literature table, and provide a detailed summary of one selected paper.

Course Learning Outcome (CLO): CLO3: Apply fundamental cryptographic techniques and simulate basic cyber-attack scenarios to evaluate system resilience

2. Learning Objectives

Upon successful completion of this assignment, students should be able to:

1. Explain the purpose, architecture, major features, and operational principles of a cybersecurity tool.
2. Install, configure, and simulate the tool in an appropriate laboratory environment.
3. Demonstrate at least three strengths and three weaknesses using practical evidence.
4. Identify, evaluate, and reference ten relevant research papers involving the selected tool.
5. Summarize a research paper, including its objectives, methodology, findings, and limitations.
6. Relate practical tool capabilities to cybersecurity problems and research applications.
7. Document experimental procedures, results, and conclusions in a reproducible technical report.

3. Selection of a Cybersecurity Tool

Each group must select one tool from a relevant cybersecurity domain. The following Table 1 shows examples, not an exhaustive list.

Table 1: List of Example Tools with corresponding Cybersecurity Doman and possible application

Cybersecurity domain	Example tools	Possible application
Packet sniffing and network analysis	Wireshark, tcpdump, TShark	Capture and inspect network traffic
Network discovery and scanning	Nmap, Masscan	Identify hosts, ports, and services
Intrusion detection and network monitoring	Snort, Suricata, Zeek	Detect suspicious traffic and analyze network activity
Digital forensics	Autopsy, Volatility, The Sleuth Kit	Examine digital evidence and memory artifacts
Web application security testing	OWASP ZAP, Burp Suite Community Edition	Inspect requests and test web application security
Vulnerability assessment	Greenbone/OpenVAS, Nessus Essentials	Identify and assess security weaknesses
Password security auditing	John the Ripper, Hashcat	Evaluate password strength using authorized test data
Malware analysis	Ghidra, REMnux tools	Analyze suspicious files in a safe lab
Wireless security analysis	Kismet, Aircrack-ng	Examine wireless networks in an authorized test environment
Security log analysis	Wazuh, Splunk	Investigate logs, alerts, and security events

Important selection requirements

- Each group must choose one primary tool and obtain instructor/TA approval before starting.
- Groups should avoid selecting the same tool unless explicitly permitted.
- The selected tool must have sufficient functionality for a practical demonstration and sufficient published research for the literature review.
- Students must verify that they can access the tool, documentation, and research papers before finalizing their selection.
- Where a commercial tool is selected, the group must clearly state any trial, licensing, or feature limitations.

Task 1: Tool Identification and Technical Study

Introduce the selected tool and explain its relevance to cybersecurity.

Your report must include:

1. Tool name, version, developer or maintainer, official website, and license or licensing model.
2. Primary purpose and cybersecurity domain.
3. Security problem or problems addressed by the tool.
4. Main features and capabilities.
5. Operating systems and hardware/software requirements.
6. Architecture and operational workflow, supported by a diagram where appropriate.
7. Installation and configuration procedure.
8. Inputs, outputs, reports, alerts, or artifacts generated by the tool.
9. Comparison with at least one alternative tool.
10. A brief explanation of why your group selected this tool.

For a packet-sniffing tool, for example, discuss packet capture, protocol dissection, filtering, traffic inspection, and relevant limitations. For other tools, adapt these items to the tool's actual functionality.

Task 2: Complete Practical Simulation and Demonstration

Each group must implement a working lab scenario that demonstrates the principal functionality of its selected tool.

A simulation must involve actual tool operation and observable results. A presentation containing only screenshots from the internet, copied commands, or a theoretical description will not qualify as a complete simulation.

5.1 Lab setup

Document:

- The operating system, virtual machines, containers, or other lab components used.
- The tool version and configuration.
- The network topology or experimental environment, where applicable.
- The test data, traffic, files, or events used.
- The commands, configuration settings, and execution steps.
- Any safety precautions and authorization boundaries.

Students should use isolated virtual machines, synthetic data, or systems they own.

5.2 Functional demonstration

Demonstrate the principal features of the tool through a realistic, reproducible scenario. Include:

- The objective of the experiment.
- The setup and configuration.
- The execution procedure.
- Evidence of successful operation.
- Interpretation of the observed output.
- A brief discussion of the result and its cybersecurity relevance.

For a packet-sniffing tool, an example scenario is to generate benign traffic between two lab systems, capture the traffic, apply protocol or display filters, inspect selected packets, and explain the results. Students should also discuss encryption and why encrypted application traffic generally prevents a passive observer from reading its protected contents.

For a vulnerability scanner, the equivalent demonstration might involve scanning an intentionally vulnerable lab machine, interpreting findings, and validating selected results. Other tools should use a scenario appropriate to their function.

5.3 Required evidence

Include the following in the report:

1. Screenshots showing the tool's configuration and execution.
2. Relevant command-line output, packet captures, alerts, logs, or generated reports.
3. Explanations accompanying the evidence.
4. A short demonstration video or live demonstration, as directed by the instructor.
5. A clear distinction between results produced by your group and results taken from external sources.

All group members must understand the complete workflow and be prepared to answer technical questions about the demonstration.

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6. Task 3: Critical Evaluation — Three Strengths and Three Weaknesses

Identify **at least three strengths and at least three weaknesses** of the selected tool. Each point must be explained and supported by practical evidence.

6.1 Strengths

For each strength, provide:

- A clear description of the capability or advantage.
- The conditions under which it is useful.
- A practical demonstration or other credible supporting evidence.
- An explanation of its significance for cybersecurity practitioners.

6.2 Weaknesses and limitations

For each weakness, provide:

- A clear description of the limitation.
- The conditions under which the limitation occurs.
- A practical demonstration, test result, or credible technical justification.
- The potential effect on security operations or investigations.
- A possible workaround, mitigation, or complementary tool, where appropriate.

6.3 Evaluation Table

Use the following table in your report. Add rows if you identify more than the minimum number of strengths and weaknesses.

No.	Type	Description	Evidence or test performed	Observed result	Practical implication
1	Strength	[Description]	[Test/evidence]	[Result]	[Implication]
2	Strength	[Description]	[Test/evidence]	[Result]	[Implication]
3	Strength	[Description]	[Test/evidence]	[Result]	[Implication]
4	Weakness	[Description]	[Test/evidence]	[Result]	[Implication]
5	Weakness	[Description]	[Test/evidence]	[Result]	[Implication]
6	Weakness	[Description]	[Test/evidence]	[Result]	[Implication]

Important: Do not merely state that a tool is easy to use, slow, expensive, or limited. Explain the relevant conditions and provide evidence. A feature that is a strength in one environment may be a limitation in another.

Task 4: Research Literature Review (Ten Papers

Each group must identify **ten distinct research papers that mention, evaluate, apply, or otherwise explicitly discuss the selected tool**.

Literature search should establish how the tool has been used in academic or applied research, rather than simply collecting papers that mention the tool in passing.

7.1 Sources for finding papers

Students may use Google Scholar, IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, Wiley Online Library, and /or Relevant scholarly journals and conference proceedings

Note: Official documentation and general websites may help explain the tool, but they do not count as research papers for the required ten-paper table.

7.2 Search strategy

Use combinations of the tool's name with relevant cybersecurity terms. For example:

- "Wireshark" AND cybersecurity
- "Wireshark" AND intrusion detection
- "Wireshark" AND network traffic analysis
- "tcpdump" AND network security
- "[Tool Name]" AND evaluation
- "[Tool Name]" AND "[relevant application]"

Modify the search terms according to the selected tool. Record the databases searched, search terms, and the date of the search.

7.3 Paper selection criteria

Each paper must:

1. Be a distinct scholarly research publication.
2. Have verifiable bibliographic information.
3. Explicitly mention or use the selected tool in a relevant technical or research context.
4. Provide enough accessible information to establish its relevance.
5. Be traceable through a DOI, publisher page, proceedings page, or another reliable scholarly record where available.

Prioritize peer-reviewed journal articles and conference papers. If the selected tool has limited academic coverage, consult the instructor/TA before including other publication types. Do not invent references, DOIs, results, or author details. Do not count the same paper twice because it appears in different databases.

7.4 Required literature Table

Prepare a table containing all ten papers with the following columns:

No.	Authors and year	Paper title	Journal/conference or publisher	How the tool is used	Key contribution or finding	DOI or stable URL
1	[Authors, year]	[Full title]	[Publication]	[Specific use]	[Main contribution]	[DOI/URL]
2	[Authors, year]	[Full title]	[Publication]	[Specific use]	[Main contribution]	[DOI/URL]
3	[Authors, year]	[Full title]	[Publication]	[Specific use]	[Main contribution]	[DOI/URL]
4	[Authors, year]	[Full title]	[Publication]	[Specific use]	[Main contribution]	[DOI/URL]
5	[Authors, year]	[Full title]	[Publication]	[Specific use]	[Main contribution]	[DOI/URL]
6	[Authors, year]	[Full title]	[Publication]	[Specific use]	[Main contribution]	[DOI/URL]
7	[Authors, year]	[Full title]	[Publication]	[Specific use]	[Main contribution]	[DOI/URL]
8	[Authors, year]	[Full title]	[Publication]	[Specific use]	[Main contribution]	[DOI/URL]
9	[Authors, year]	[Full title]	[Publication]	[Specific use]	[Main contribution]	[DOI/URL]
10	[Authors, year]	[Full title]	[Publication]	[Specific use]	[Main contribution]	[DOI/URL]

The “How the tool is used” column must explain the tool's actual role in each paper. For example, it may have been used to capture traffic, analyze protocols, generate or inspect data, validate an experiment, or evaluate a security solution.

NOTE: *Do not copy abstracts into the table. Summarize each paper's relevance in your own words.*

Task 5: Detailed Summary of One Research Paper

Select one of the ten papers and prepare a detailed critical summary. Choose a paper that provides meaningful insight into the selected tool's application, capabilities, evaluation, or limitations.

Your summary must cover the following subsections.

8.1 Paper identification

- Complete reference in the required citation style.
- DOI or stable publisher URL.
- Research domain and application.

8.2 Research problem and motivation

Explain the problem addressed by the authors, why it matters, and what gap or need motivated the study.

8.3 Research objectives and questions

State the main objectives or research questions, based on the paper.

8.4 Methodology

Describe the research design, experimental setup, datasets, participants, test environment, or analytical method, as applicable. Explain precisely how the selected tool was used.

8.5 Key findings

Summarize the principal findings, reported results, measurements, or conclusions. Include numerical results only when they are actually reported in the paper.

8.6 Role of the selected tool

Explain what the tool contributed to the study and whether it was central to the research or used as a supporting instrument.

8.7 Limitations and critical assessment

Discuss the limitations acknowledged by the authors and any additional limitations that you can reasonably identify from the study design or evidence. Clearly distinguish the authors' statements from your own evaluation.

8.8 Lessons for practitioners

Explain what a cybersecurity professional could learn from the paper and how its findings relate to your group's practical demonstration.

Suggested length: 800–1,200 words, excluding the reference.

Task 6: Comparative Discussion and Conclusions

Conclude the assignment by integrating your practical and research findings.

Address the following questions:

1. Which cybersecurity problems is the selected tool best suited to address?
2. Which of its strengths were confirmed by your experiments?
3. Which weaknesses were observed, and which could not be tested directly?

4. How does literature demonstrate the tool's practical or research value?
5. How does the tool compare with the alternative tool identified in Task 1?
6. What additional tools, safeguards, or procedures would complement it?
7. What improvements or future research directions would you recommend?

Your conclusions must be based on your experimental evidence and reviewed literature.

10. Group Work and Individual Accountability

Each group may have a maximum of three students. All members must contribute meaningfully to the assignment and understand its major components.

The report must include a contribution statement identifying each member's responsibilities. For example:

- **Member 1:** Tool setup, architecture, and simulation.
- **Member 2:** Strengths/weaknesses experiments and evidence.
- **Member 3:** Literature search, research-paper table, and paper summary.

These roles are illustrative; they do not exempt any member from understanding the overall work. The group may distribute tasks differently, but every member must be able to explain the tool, the experimental results, and the principal conclusions during questioning.

11. Submission Requirements

Submit the following deliverables:

1. **Technical report:** PDF format, containing all required tasks and references.
2. **Experimental artifacts:** Relevant configuration files, scripts, sample data, logs, packet captures, or reports, where appropriate and safe to share.
3. **Evidence:** Original screenshots and a live demonstration.
4. **Literature table:** All ten papers, with complete and verifiable references.
5. **Research paper summary:** Included as a dedicated report section.
6. **Contribution statement:** A brief account of each member's work.

Recommended report structure

1. Title page and group member details
2. Abstract or executive summary
3. Introduction and objectives
4. Tool overview and technical background
5. Lab setup and methodology
6. Simulation and functional demonstration

7. Strengths and weaknesses evaluation
8. Literature search strategy
9. Ten-paper literature table
10. Detailed summary of one paper
11. Comparative discussion
12. Conclusions and future recommendations
13. References
14. Appendices: additional evidence and experimental artifacts

Formatting: Use a consistent academic citation style, such as IEEE. Number figures and tables, cite them in the text, and provide appropriate captions. Ensure that all external sources, diagrams, and adapted materials are acknowledged.

12. Academic Integrity and Safe Use

- All experiments must be conducted on systems and data that the group owns or is authorized to test.
- Do not capture other people's private communications or credentials.
- Do not scan, exploit, disrupt, or intercept traffic on public or institutional systems without explicit authorization.
- Use synthetic data when appropriate.
- Cite all external sources and identify any use of generative AI in accordance with the course's academic integrity policy.
- Fabricated experiments, screenshots, research papers, DOIs, or results are unacceptable.

13. Assessment Rubric

The assignment will be evaluated out of 100 marks.

Assessment component	Marks	Evaluation criteria
Tool identification and technical understanding	10	Accuracy, relevance, technical depth, and understanding of the tool
Lab setup and simulation	20	Working implementation, completeness, reproducibility, and quality of evidence
Strengths and weaknesses	10	At least three of each, supported by meaningful experiments or evidence
Ten-paper literature review	20	Relevance, authenticity, research quality, correct referencing, and depth of analysis

Detailed summary of one paper	10	Understanding of problem, methodology, findings, tool usage, and limitations
Comparative discussion and conclusions	5	Evidence-based synthesis and practical recommendations
Report quality and academic integrity	5	Organization, clarity, citations, presentation of evidence, and originality
Individual viva and contribution	20	Individual understanding, ability to explain the work, and demonstrated contribution

Performance expectations

- **Excellent:** Complete working simulation; convincing evidence for the strengths and weaknesses; ten authentic, relevant papers; insightful paper summary; and strong individual understanding.
- **Good:** Functional simulation and sound analysis, with minor gaps in evidence, literature depth, or critical discussion.
- **Satisfactory:** Basic tool operation is demonstrated, but analysis is limited or some required components are incomplete.
- **Needs improvement:** Predominantly theoretical work, insufficient practical evidence, weak or unverifiable references, or limited understanding during the viva.

14. Final Checklist

Before submission, verify that your group has:

- Selected and obtained approval for one cybersecurity tool.
- Explained its purpose, architecture, features, and installation.
- Demonstrated its principal functionality in a controlled lab.
- Presented original evidence of the simulation.
- Identified and demonstrated at least three strengths.
- Identified and demonstrated at least three weaknesses.
- Identified ten distinct and verifiable research papers.
- Completed the literature table with references and stable links.
- Written an 800–1,200-word summary of one selected paper.
- Compared practical findings with research literature.
- Included individual contributions and followed safe-use requirements.
- Checked citations, formatting, and academic integrity.

The central expectation is that students demonstrate, evaluate, and critically understand a real cybersecurity tool, not merely describe it.